

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Given

- **Training MSE = 0.85**
- **Testing MSE = 5.20**

The training error is very low, while the testing error is much higher. This indicates that the model performs well on the training data but fails to generalize to unseen data. This is a classic case of **overfitting**.

1. Overfitting

- The neural network has memorized the training data instead of learning general patterns.
- As a result, it performs poorly on new (testing) data.

2. Insufficient Training Data

- A small dataset may not represent all possible student score patterns.
- The model cannot learn enough variations.

3. Complex Model

- Too many hidden layers or neurons increase model complexity.
- This leads to memorization rather than learning.

4. Data Distribution Difference

- Training and testing datasets may have different characteristics.
- The model cannot adapt well to unseen data.

5. Lack of Regularization

- No dropout, L1/L2 regularization, or early stopping allows the model to overfit.

• Increase Training Data

- Collect more student records or use data augmentation where applicable.

• Apply Regularization

- Use L1/L2 regularization or Dropout layers to reduce overfitting.

• Early Stopping

- Stop training when validation loss starts increasing.

• Simplify the Model

- Reduce the number of hidden layers or neurons.

• Cross-Validation

- Use k-fold cross-validation to evaluate model performance more reliably.

• Proper Data Splitting

- Ensure training and testing datasets come from the same distribution and are randomly divided.

Results

- The low **training MSE (0.85)** shows that the model has learned the training data well.
- The high **testing MSE (5.20)** indicates poor generalization.
- The large gap between training and testing errors confirms **overfitting**.
- Applying regularization, increasing training data, simplifying the model, and using early stopping can reduce the testing error and improve overall performance.

Conclusion

The neural network suffers from **overfitting**, resulting in excellent training performance but poor testing performance. By using techniques such as regularization, early stopping, increasing training data, simplifying the model, and cross-validation, the model can generalize better and achieve lower testing MSE. This addresses the required error identification, reasoning, correction strategy, and interpretation expected in the assessment rubric.

Code of Conduct:

*I **P.AKHILA(192425225)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Akhila

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided